

The empirical recurrence for OEIS A304132, recovered from its tail

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Abstract

OEIS A304132 records “Empirical recurrence of order 79” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 437$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 9$, so the recurrence is proved for every $n \geq 88$. Its last coefficient is $c_{79} = 36 \neq 0$, so no further tail satisfies anything shorter; any order-79 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A304132 is “Number of $n \times 7$ 0..1 arrays with every element unequal to 0, 1, 3 or 6 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

21, 103, 89, 244, 582, 1321, 2945, 6333, ...

The entry states:

Empirical recurrence of order 79 (see link above)

As of the “Last modified” line on the live entry (May 07 2018 08:03:30, revision 8) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 437$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 437$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 79: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 9$ is the first whose minimal order is exactly the stated 79.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 874$ exact terms from $n = 9$ on, it returns order 79 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{79} c_i a(n-i),$$

c_1	1	c_{21}	3664738	c_{41}	822622843	c_{61}	37286349
c_2	0	c_{22}	3783356	c_{42}	935755432	c_{62}	3664622
c_3	39	c_{23}	11270138	c_{43}	256585817	c_{63}	−16369048
c_4	−30	c_{24}	−5942394	c_{44}	−772601263	c_{64}	−10252900
c_5	4	c_{25}	−13754718	c_{45}	−1006585933	c_{65}	−1722997
c_6	−708	c_{26}	−40587025	c_{46}	−400567110	c_{66}	3651113
c_7	380	c_{27}	−1951162	c_{47}	571030807	c_{67}	2072817
c_8	−184	c_{28}	36049328	c_{48}	870807889	c_{68}	437560
c_9	7909	c_{29}	115383004	c_{49}	431627520	c_{69}	−606943
c_{10}	−2380	c_{30}	47386592	c_{50}	−323415770	c_{70}	−295877
c_{11}	3786	c_{31}	−68110167	c_{51}	−608262806	c_{71}	−67377
c_{12}	−60450	c_{32}	−261753180	c_{52}	−347746144	c_{72}	70450
c_{13}	3404	c_{33}	−171951198	c_{53}	132536055	c_{73}	27882
c_{14}	−46932	c_{34}	86155050	c_{54}	343200153	c_{74}	5916
c_{15}	331361	c_{35}	476779011	c_{55}	214824173	c_{75}	−5060
c_{16}	69907	c_{36}	400314980	c_{56}	−33147862	c_{76}	−1530
c_{17}	396017	c_{37}	−46911324	c_{57}	−156033119	c_{77}	−228
c_{18}	−1317076	c_{38}	−698804197	c_{58}	−102270041	c_{78}	168
c_{19}	−706470	c_{39}	−692917870	c_{59}	446221	c_{79}	36
c_{20}	−2429001	c_{40}	−77867329	c_{60}	56812960		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 88$, and it is the recurrence OEIS A304132 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 9$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{79} - c_1 x^{79-1} - \dots - c_{79}$. Its constant term is $-c_{79} = -36$, which is not zero, so $q_{\min}(0) \neq 0$ and

$q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 79 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 79, so $\deg q = 79$, and it is monic. It holds from some index t on. From $\max(t, 9)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 79, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 28 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 79, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 36.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-79 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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